

Lab 2 – TasteBuddies Product Specification

Oronde Brown

Old Dominion University

CS411W

Professor Hosni

March 7, 2025

Version 1

Table of Contents

1	Introduction.....	3
1.1	Purpose.....	3
1.2	Scope.....	3
1.3	Definitions, Acronyms, and Abbreviations	4
1.4	References.....	6
1.5	Overview.....	8
2	Overall Description.....	8
2.1	Product Perspective.....	8
2.2	Product Functions	10
2.3	User Characteristics	12
2.4	Constraints	12
2.5	Assumptions and Dependencies	13

List of Figures

Figure 1	9
----------------	---

List of Tables

Table 1	10
---------------	----

1 Introduction

1.1 Purpose

This SRS document is presenting the details of the prototype for a web application called TasteBuddies. The document will give a detailed description of what the app is and how the prototype is being developed. The document's overall goal is to give insight as to how the app prototype will function and the technical details behind it.

1.2 Scope

TasteBuddies is an application that is designed to help make the dining experience the best it can be by finding the food that is right for an individual's taste. By using and interacting with the app, Tastebuddy's, which are one of the user roles, will continuously give Tastebuddies information to better their taste matching algorithm, and in return be able to receive better recommendations. TasteBuddy's create a taste profile upon account creation which allows them to set their food preferences and indicate dishes they already enjoy, enabling recommendations that are continuously improved over time through machine learning. The filters for the taste profile will also include aspects like allergies, dietary restrictions, and the user's preferred dining experience.

TasteBuddies will allow for live updates which will allow tastebuddys to know exactly what is going on in the various restaurants in terms of specials, reviews, wait times, and operating hours. This is done through crowdsourcing data from other users. The app is will also be integrated with the Google API, which will allow tastebuddys to see the activity level in each restaurant to aid tastebuddys who prefer to avoid loud and crowded establishments. The app will also have systems in place similar to the taste matching algorithm that will help in prioritizing reviews from other tastebuddies as they have similar tastes. The app will also feature group

dining which will allow for dish recommendations based on the taste profiles of every individual in a group.

1.3 Definitions, Acronyms, and Abbreviations

Crowdsourced Data: User-generated data on restaurant wait times, dish availability, and quality, among others.

Curated Reviews: Reviews presented and weighted based on users with similar Taste Profiles.

Daily Dish Report: Provides live updates from TasteBuddies and restaurants such as new reviews, specials, and dishes.

Data clustering: grouping diners in a group that is more similar to determine taste profiles and recommendations

Dining Filters: Ability to filter restaurants by location, cuisine, occasion, and how busy they are.

Generic reviews: The issue of unauthentic online reviews, which the app addresses by focusing on personalized recommendations.

Google API: An external tool integrated into the app that provides real-time data on how busy a restaurant is.

Group Dining Algorithm: Algorithm that combines multiple users profiles and provides reviews for restaurants and dishes that best match the group preferences.

Group Indecision: Conflicting opinions and preferences of a group lead to more difficult decision making which causes delays.

High financial risk: The risk of losing/wasting money based on a decision.

ODU: Old Dominion University.

Overwhelming choice: An excessive number of options to choose from which makes decisions difficult.

Recommendation Algorithm: Algorithm that provides users with relevant recommendations based on their matched TasteBuddies, taste profile, and interacted content.

Restaurants: Venue that provides a sit-down dining experience where primary revenue is prepared food. It must have a nice bathroom.

Safe space: Space where people are free to express and enjoy their interest without fear of being judged.

Social engagement: Promote users to interact with one another and be involved within the community.

Super TasteBuddies: Taste influencers or food experts that have specialized knowledge and can recommend specific cuisines or dishes.

Tailored Recommendations: Personalized recommendations based on a user's taste profile.

TasteBuddies: Users with highly similar taste profiles which lead to improved recommendations based on aligned tastes.

Taste Matching Algorithm: A key Algorithm of the app that pairs users based on similar taste profiles.

Taste Profiles: Personalized profiles created by each user based on their taste preferences, such as preferences for spicy, sweet, salty, etc.

1.4 References

Beaton, C., “Why You Can’t Really Trust Negative Online Reviews,” *Nytimes.com*, Jun. 14, 2018. [Online]. Available: <https://www.nytimes.com/2018/06/13/smarter-living/trust-negative-product-reviews.html#>

Dunbar, R. I. M., “Breaking Bread: the Functions of Social Eating,” *Adaptive Human Behavior and Physiology*, vol. 3, no. 3, pp. 198–211, Mar. 2017, doi: <https://doi.org/10.1007/s40750-017-0061-4>.

Escoffier, Auguste School of Culinary Arts, “2024 Consumer Dining Trends: How Americans Are Spending on Restaurants and Takeout,” *Escoffier*, Sep. 25, 2024. [Online]. Available: <https://www.escoffier.edu/blog/world-food-drink/consumer-dining-trend-statistics/>

Food price inflation over 2016–2020 equal to economy-wide inflation, www.ers.usda.gov, Feb. 14, 2024. [Online]. Available: <https://www.ers.usda.gov/data-products/chart-gallery/gallery/chart-detail/?chartId=58350>

Grieve, P., “Why Eating at Restaurants Is So Expensive Right Now,” *Money*, May 20, 2024. [Online]. Available: <https://money.com/restaurant-prices-increase-inflation/>

Harris poll, “US Adults’ Eating Habits Survey Results,” Nov. 22, 2017. [Online]. Available: <https://drive.google.com/file/d/1Hvq3SZbYq6AsSoSGVFpZ8O52rW2TY6UW/view> (accessed Oct. 04, 2024).

Higgs, S. and Thomas, J., “Social Influences on Eating,” *Current Opinion in Behavioral Sciences*, vol. 9, pp. 1–6, Oct. 2015. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S235215461500131X>

Home Run Inn Pizza, “Culinary Confessions: Cooking Habits of Gen Z & Millennials,” Home Run Inn Pizza, Aug. 16, 2023. [Online]. Available:

<https://www.homeruninnpizza.com/news-blog/fun-facts/cooking-habits-gen-z-millennials/>

Mohanty, M., “Impulsive Buying in the F&B Industry: A new opportunity for restaurants?,”

idsnext, Jul. 06, 2023. [Online]. Available: <https://idsnext.com/blogs/impulsive-buying-in-the-fb-industry-a-new-opportunity-for-restaurants/> (accessed Oct. 04, 2024).

OpenTable, “Restaurant Impact Report,” OpenTable, Oct. 2023. [Online]. Available:

https://www.opentable.co.uk/c/wp-content/uploads/sites/342/2023/10/opentable_uk-restaurant-impact-report-2023.pdf

Perkins, T., “Your food is more expensive – are US corporate profits to blame?,” The Guardian, Jul. 26, 2024. [Online]. Available:

<https://www.theguardian.com/environment/article/2024/jul/26/food-price-inflation-corporate-profit>

Statista Research Department, “Frequency of eating at restaurants in the US 2022,” Statista, Nov.

15, 2023. [Online]. Available: <https://www.statista.com/statistics/1324709/frequency-of-eating-out-at-restaurant-in-the-us/>

Thurnell-Read, T., “Open arms: the role of pubs in tackling loneliness,” Loughborough

University, Jan. 2021. [Online]. Available: <https://hdl.handle.net/2134/13663715.v1>

U.S. Travel Association, “U.S. Travel and Tourism Overview (2019),” U.S. Travel Association,

Mar. 2020. [Online]. Available:

https://www.ustravel.org/system/files/media_root/document/Research_Fact-Sheet_US-Travel-and-Tourism-Overview.pdf

Vaghasiya, K., “15 Fake Review Statistics You Can’t Ignore (2024),” WiserNotify. [Online]. Available: <https://wisernotify.com/blog/fake-review-stats/#combating-fake-reviews:-strategies-and-tools> (accessed Oct. 04, 2024).

Wunsch, N.-G., “Average annual food away from home expenditures of United States households from 2010 to 2022,” Statista, Feb. 12, 2024. [Online]. Available: <https://www.statista.com/statistics/237215/average-away-from-home-food-expenditures-of-united-states-households/>

University of Oxford, “Social eating connects communities,” University of Oxford, Mar. 16, 2017. [Online]. Available: <https://www.ox.ac.uk/news/2017-03-16-social-eating-connects-communities>

1.5 Overview

This product specification document will give insight into the system architecture, assumptions dependencies, hardware, and software aspects of the TasteBuddies prototype. In the coming sections, descriptions for the plans of the prototype's implementation will be presented.

2 Overall Description

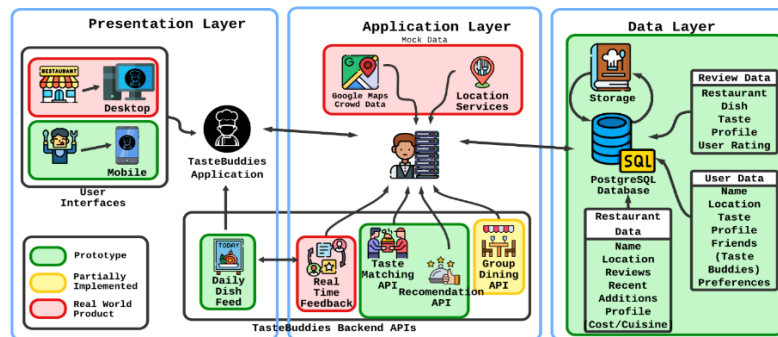
2.1 Product Perspective

The TasteBuddies prototype will be developed using flask and jinja2. SQLite will also be used to store data for the users which will be implemented with the help of mock data. The

TasteBuddies prototype will be deployed on a digital ocean server. The system behind TasteBuddies prototype uses a three-tier architecture as shown in Figure 1.

Figure 1

TasteBuddies Prototype Major Functional Component Diagram



The first layer is the presentation layer, which users such as tastebuddys and restaurant managers interact with. Tastebuddy interaction will involve aspects like getting recommendations, looking at the feed, and getting information regarding restaurants. For restaurant managers, they will interact by updating their information such as specials, dishes, operating hours, and other information customers would need.

The next layer of Tastebuddies is the application layer. Key features within the layer include location services, which ensure users are getting recommendations that are near them. This will be implemented through mock data in the prototype. Additionally, the application layer incorporates three APIs that TasteBuddies uses will use to function. The taste-matching API is how TasteBuddies can recommend users to each other if they have similar taste profiles. The recommendation API is what matches the users to the different dishes, specials, and restaurants based on their own taste profiles. The final Api is the group dining API, which, as the name

suggests, allows TasteBuddies to combine the preferences of many in a group and recommend a restaurant based on everyone's taste profiles.

The final layer is the data layer, where all the data of users, restaurants, and other aspects of the app are stored. A PostgreSQL database is used to keep all the information coming into the app stored and organized. This layer functions by taking user-inputted information, such as a user's taste profile, and passing this information through the webserver to the user profile so it can be saved in the PostgreSQL database. This information will then be pulled anytime the APIs need to use it.

2.2 Product Functions

The prototype will be implementing the core features necessary to give users an idea of what the authentic experience of the real-world product will look like. The prototype will be centered around the customer user role. In the prototype, Customers will have access to account creation, adding/finding friends, a daily dish feed, taste matching, taste profiles, and other personalization type features. Table 1 shows a better breakdown of how each feature will be implemented and shows that hard coding and mock data will help with simulating aspects needed like other users, restaurants, reviews, and other data needed to help the app function.

Table 1

Table of TasteBuddies Prototype Features

Category		Features	Prototype	Additional Notes
Expanded User Mobile	Social Engagement	TasteBuddies	Fully implemented	
		Super TasteBuddies	Partially implemented	Hard coded
		Add/Find Buddies	Fully implemented	

App Features		Follow TasteBuddy	Eliminated	
		Follow Restaurant	Eliminated	
		Add Kudos	Eliminated	
	Live Interactive Updates	Daily Dish feed	Fully implemented	
		Add reviews	Partially implemented	Mock data provided
		Post restaurant update	Eliminated	
		Post dish update	Eliminated	
		Notifications	Eliminated	
	Recommendation Engine	Taste Profile	Fully implemented	
		Read Reviews	Partially Implemented	
		Taste Matching	Fully implemented	
		Dish Recommendation	Fully implemented	
		Group Restaurant Matching	Partially Implemented	Implementation time dependent
		Rewards	Eliminated	
		Adaptive Taste Profile personalization	Eliminated	Need active data over time
	Search	Restaurant filtering	Eliminated	
		Dish filtering	Eliminated	
	Engagement features	Rewards	Eliminated	
		Badges	Eliminated	
		Challenges	Eliminated	

The features that will be fully implemented in the prototype are account creation, tastebuddies, adding/finding friends, daily dish feed, taste profiles, taste matching, and dish recommendations. Account creation gets everything started as part of creating the account is setting up your taste profile. The taste profile will allow the system to get a sense of the user's taste and then run them through the taste matching algorithm to bring the user dish recommendations based on the user's taste profile. It will also allow them to find tastebuddies which are the other users that have similar tastes and see their daily dish which shows featured and recommended dishes from restaurants.

Features that will be partially implemented will be super tastebuddies, add reviews, read reviews, and group restaurant matching. The features super tastebuddies, add reviews, and read reviews are partially implemented as we will be using hard coding and data to simulate how they would work. As for group taste matching we could see it being fully implemented if time allows. Group matching would allow dishes to be recommended based off everyone in a specific groups taste profiles which will better help in finding restaraunts. The rest of the features will be eliminated in the prototype as they are mainly quality of life features that are not essential in showing the capabilities of TasteBuddies.

2.3 User Characteristics

The user roles included in the prototype will be that of the tastebuddy and the administrator. The role of the tastebuddy is to be the users who create the accounts, get the taste profile, get recommended dishes, and other things of that nature. The tastebuddy's are the focus of the prototype and are the ones that will make up much of the real-world product. The expertise level for this role would be basic and anyone who has had a smart phone app before should have no problems using the app. The other role included in the prototype is the administrator. This role is more for ensuring that system security is intact, accessing analytical data about preferences and checking patterns, and making sure the system data is backed up. The expertise for this role is a little more advanced as the administrator will have to have knowledge of both cybersecurity and on how to analyze data. It is also important to mention the role of restaurant which will be left out in the prototype but will be made up for with the help of mock data.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

For the purposes of the prototype TasteBuddies will use three APIs, which are the taste matching API, group dining API, and recommendation API. Software frameworks that will be used are React, Flask with Jinja2, and SQL Alchemy. As for the languages it will be developed using Python, HTML, CSS, and JavaScript.